

Teen Mental Health Analysis Using Python

Data Visualization Project

Technologies: Python | Pandas | Matplotlib | Seaborn

Introduction

Teen mental health has become an important area of study due to the increasing influence of social media, academic pressure, screen time, and changing lifestyles. Understanding how these factors relate to stress, anxiety, sleep quality, physical activity, and addiction can help researchers, educators, and policymakers make informed decisions.

This project performs Exploratory Data Analysis (EDA) on a Teen Mental Health dataset using Python. Pandas is used for data loading and preprocessing, while Matplotlib and Seaborn are used to create informative visualizations. The objective is to convert raw data into meaningful visual insights that help understand behavioural and mental health patterns among teenagers.

Dataset Description

The dataset contains 1,200 observations and 12 attributes describing demographic, behavioural, academic, and mental health characteristics of teenagers.

Features include:

- Age
- Gender
- Daily Social Media Hours
- Platform Usage
- Sleep Hours
- Screen Time Before Sleep
- Academic Performance
- Physical Activity
- Social Interaction Level
- Stress Level
- Anxiety Level
- Addiction Level

Libraries Used

- Pandas – Data loading and preprocessing
- Matplotlib – Basic charts and customization
- Seaborn – Statistical visualizations including count plots, regression plots and heatmaps

1. Gender Distribution (Pie Chart)

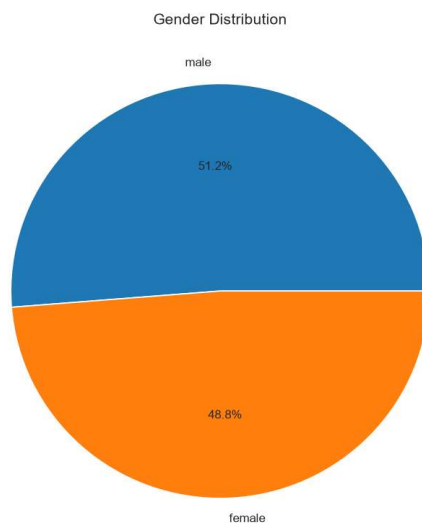
Description

The Gender Distribution pie chart provides a visual representation of the proportion of male and female participants included in the Teen Mental Health dataset. Examining the gender composition is an essential step in Exploratory Data Analysis (EDA) because it helps determine whether the dataset is balanced and representative. A balanced dataset ensures that any conclusions drawn from subsequent analyses are less likely to be influenced by sampling bias.

Code

```
plt.figure(figsize=(6,6))
df['gender'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title('Gender Distribution')
plt.ylabel('')
plt.show()
```

Graph



Interpretation

The pie chart indicates the percentage of male and female students represented in the dataset. The proportions suggest that both genders are well represented, allowing meaningful comparative analysis in later stages of the project. A balanced gender distribution increases the credibility of statistical observations and minimizes the possibility of gender bias affecting the overall results. This visualization establishes the demographic structure of the dataset and confirms that it is suitable for analyzing behavioural and mental health characteristics across different gender groups. Future analyses may explore whether stress levels, anxiety levels, sleep duration, or social media addiction differ between male and female students.

2. Platform Usage Distribution (Count Plot)

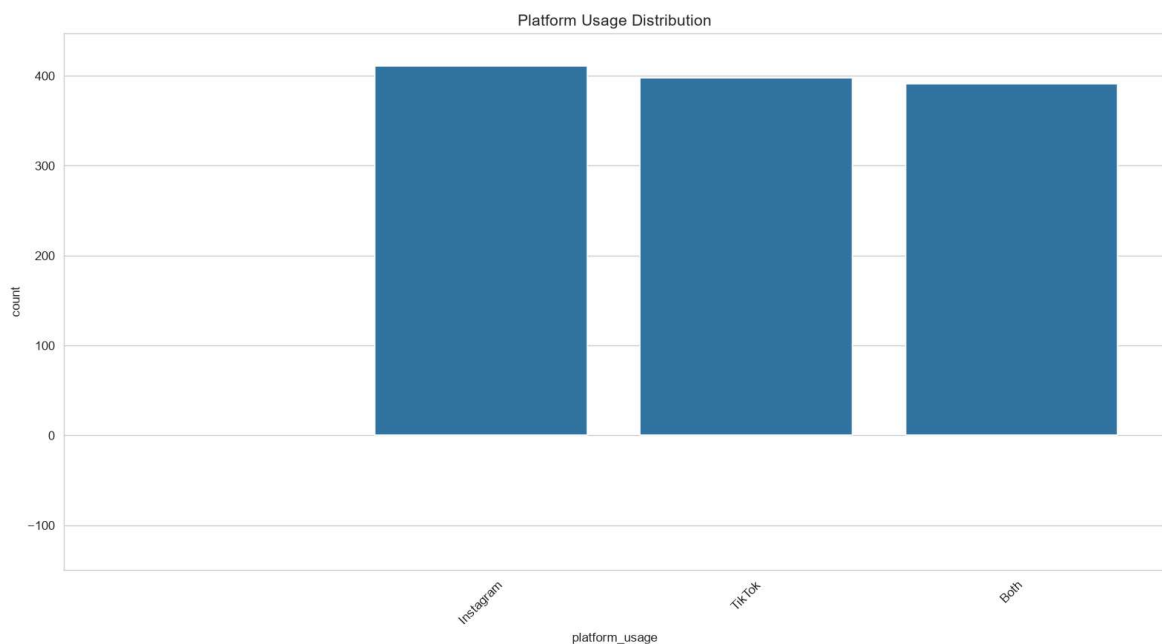
Description

The Platform Usage Distribution chart illustrates the frequency of different social media platforms used by teenagers in the dataset. Social media has become an integral part of adolescents' daily lives, influencing communication, entertainment, education, and psychological well-being. Identifying the most commonly used platforms helps provide context for understanding behavioural patterns associated with digital engagement.

Python Code

```
plt.figure(figsize=(8,5))
sns.countplot(x='platform_usage',data=df,
order=df['platform_usage'].value_counts().index)
plt.xticks(rotation=45)
plt.show()
```

Graph



Interpretation

The count plot highlights the popularity of various social media platforms among teenagers. Platforms with taller bars represent those used by a greater number of students, while shorter bars indicate comparatively lower usage. The distribution demonstrates that teenagers tend to concentrate their online activity on a few major platforms rather than using all platforms equally.

Although the visualization does not directly measure mental health outcomes, it provides valuable behavioural insights that can support future analysis. Platform preference can later be compared with variables such as addiction level, stress level, and daily social media hours to investigate whether certain usage patterns are associated with different mental health indicators.

3. Sleep Hours Distribution (Histogram)

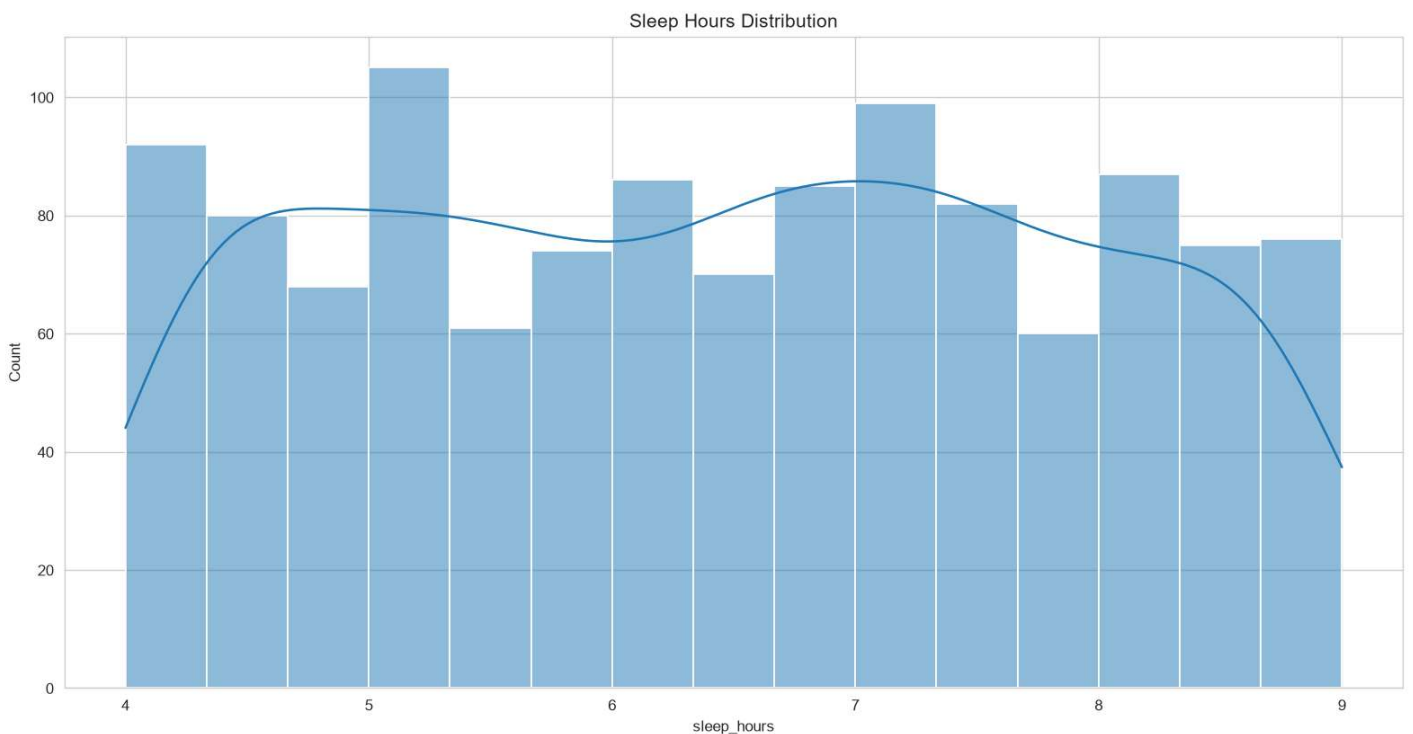
Description

The Sleep Hours Distribution histogram illustrates how daily sleep duration varies among teenagers included in the dataset. Adequate sleep plays a critical role in maintaining physical health, cognitive performance, emotional regulation, and overall mental well-being. Insufficient sleep has been associated with increased stress, anxiety, depression, reduced concentration, and lower academic performance.

Python Code

```
sns.histplot(df['sleep_hours'], bins=15, kde=True)  
plt.title('Sleep Hours Distribution')  
plt.show()
```

Graph



Interpretation

The histogram shows that most students sleep within a moderate range of hours, while fewer students appear at the extreme ends of the distribution. The KDE curve suggests the general pattern of sleeping behaviour among participants without relying solely on histogram bars. Understanding sleep distribution is important because sleep quality directly influences emotional stability, learning ability, and overall mental health. Although this visualization does not establish causal relationships, it provides valuable descriptive insights into teenage sleeping behaviour and creates opportunities for future analysis involving stress levels, anxiety, and academic performance.

5. Correlation Heatmap

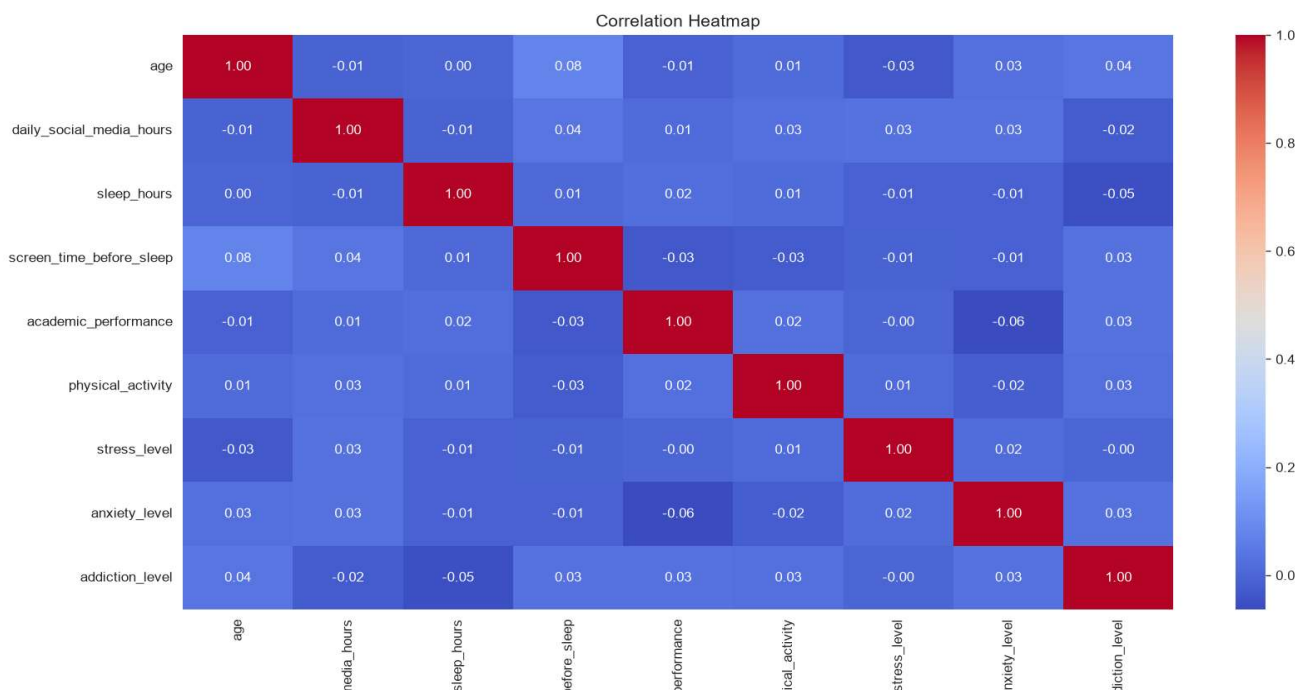
Description

The Correlation Heatmap visualizes the strength and direction of linear relationships among all numerical variables in the Teen Mental Health dataset. Correlation analysis is one of the most widely used techniques in Exploratory Data Analysis because it helps identify whether two variables increase together, decrease together, or exhibit little measurable association.

Python Code

```
plt.figure(figsize=(10,8))
sns.heatmap(df.select_dtypes(include='number').corr(),
            annot=True,cmap='coolwarm',fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```

Graph



Interpretation

The heatmap indicates that most correlation coefficients are close to zero, suggesting weak linear relationships among the variables included in this dataset. This implies that no single variable strongly explains mental health outcomes, and psychological well-being is likely influenced by multiple interacting behavioural and environmental factors. Although strong correlations are not observed, the heatmap remains an essential exploratory tool because it helps identify variables requiring further investigation. It also demonstrates that more advanced analytical methods, such as regression modelling or machine learning algorithms, may be necessary to uncover complex patterns that cannot be detected through simple correlation analysis.

6. Academic Performance vs Addiction (Regression Plot)

Description

The Academic Performance vs Addiction regression plot explores the relationship between students' academic performance and their addiction levels. Each point in the graph represents an individual student, while the fitted regression line summarizes the overall trend observed across the dataset. Regression plots are widely used in data analysis to examine whether changes in one variable are associated with changes in another.

Python Code

```
sns.regplot(  
x='academic_performance',  
y='addiction_level',  
data=df,  
scatter_kws={'alpha':0.4})  
plt.title('Academic Performance vs Addiction')  
plt.show()
```

Graph



Interpretation

The scatter points are distributed across the graph without forming a strong linear pattern, and the regression line shows only a weak overall trend. This suggests that academic performance does not exhibit a strong linear relationship with addiction level in the current dataset. The visualization indicates that academic achievement is likely influenced by multiple factors beyond digital addiction, including study habits, sleep quality, motivation, family environment, and psychological well-being. Therefore, while addiction may contribute to academic outcomes, it should not be considered the sole determining factor. More advanced statistical analysis would be required to better understand these complex relationships.

Conclusion

This project successfully demonstrates how **Python-based Exploratory Data Analysis (EDA)** can be used to transform raw data into meaningful insights about teenage mental health and behavioural patterns. By utilizing powerful Python libraries such as **Pandas, Matplotlib, and Seaborn**, the dataset was cleaned, analyzed, and visualized to better understand factors related to social media usage, sleep habits, stress levels, anxiety, addiction, and academic performance.

Through various visualizations, including the **Gender Distribution Pie Chart, Platform Usage Count Plot, Sleep Hours Histogram, Stress Level Histogram, Correlation Heatmap, and Academic Performance vs Addiction Regression Plot**, important characteristics of the dataset were explored. Each visualization provided valuable insights into the distribution of variables and the relationships between different aspects of teenagers' daily lives. The analysis revealed that while several lifestyle factors are associated with mental health, most numerical variables exhibit **weak linear correlations**, indicating that psychological well-being is influenced by multiple interacting factors rather than a single measurable variable.

The project also highlights the importance of **data visualization** in simplifying complex datasets and supporting evidence-based interpretation. Instead of relying solely on numerical statistics, graphical representations make it easier to identify trends, patterns, and potential areas for further investigation. Although this analysis does not establish cause-and-effect relationships, it provides a strong foundation for future predictive modelling and advanced machine learning applications.

Overall, this project strengthened practical skills in **data preprocessing, exploratory data analysis, statistical interpretation, and visualization using Python**. It demonstrates how analytical techniques can be applied to real-world datasets to gain meaningful insights and support informed decision-making. The knowledge and experience gained from this project provide a solid foundation for more advanced studies in **Data Science, Machine Learning, and Artificial Intelligence**, while also serving as a valuable addition to a professional portfolio.

References

1. Kaggle – Teen Mental Health Dataset (Dataset Source)
2. Python Documentation – <https://docs.python.org/>
3. Pandas Documentation – <https://pandas.pydata.org/docs/>
4. Matplotlib Documentation – <https://matplotlib.org/stable/>